

PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

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Project: Heart Disease Analysis using Tableau

Heart disease is one of the leading causes of death worldwide and continues to pose significant challenges to healthcare systems. Factors such as unhealthy lifestyles, poor dietary habits, smoking, obesity, diabetes, and physical inactivity contribute to the increasing prevalence of cardiovascular diseases. Although healthcare organizations collect large volumes of patient data, extracting meaningful insights from this information remains a challenge due to its complexity and lack of effective visualization.

This project presents **Heart Disease Analysis using Tableau**, an interactive business intelligence solution designed to transform raw healthcare data into meaningful visual insights. The project analyzes patient demographics, lifestyle habits, and medical conditions to identify factors associated with heart disease. Using Tableau's interactive dashboards, KPI cards, filters, calculated fields, and data storytelling capabilities, users can explore healthcare trends, compare patient groups, and understand disease patterns through intuitive visualizations.

The project includes **three interactive dashboards, ten analytical visualizations, KPI cards, interactive filters**, and a **Tableau Story**, all integrated into a **Flask-based web application** for improved accessibility. The web application provides users with a centralized platform to explore dashboards, review analytical findings, and access project information through an organized and user-friendly interface.

The insights generated from this project can assist healthcare professionals, researchers, and students in understanding the relationship between demographic characteristics, lifestyle factors, and medical conditions associated with heart disease. By converting healthcare data into actionable visual information, the project promotes data-driven decision-making, enhances awareness of cardiovascular risk factors, and demonstrates the practical application of business intelligence in the healthcare domain.

Scenario 1: Dr. Sharma – Cardiologist

Dr. Sharma is a senior cardiologist working at a metropolitan hospital. She wants to understand which lifestyle factors contribute most to the increase in heart disease cases among middle-aged patients. Using Tableau dashboards, she can analyze patient data segmented by age, gender, BMI, cholesterol levels, and smoking habits. This helps her identify high-risk groups and design targeted awareness campaigns, such as advising patients on weight management and smoking cessation programs.

Scenario 2: Ramesh – Healthcare Administrator

Ramesh works with a government health department and is tasked with developing preventive health policies. He uses Tableau dashboards to study trends in heart disease prevalence across different regions, comparing rural and urban populations. By analyzing correlations between sedentary lifestyle indicators and disease rates,

he can recommend policies such as fitness programs in workplaces, stricter tobacco regulations, and subsidies for healthier food options. Tableau helps him present these findings in interactive dashboards for decision-makers.

Scenario 3: Anita – Patient

Anita, a 45-year-old professional with a family history of heart disease, wants to monitor her health risks. With simplified Tableau dashboards provided by her healthcare provider, she can visualize her risk factors such as cholesterol levels, blood pressure, and lifestyle habits compared to healthy benchmarks. The dashboard highlights actionable steps like increasing physical activity or reducing fat intake. This empowers Anita to make informed decisions about her lifestyle and proactively reduce her risk of developing heart disease.

1.2 Purpose

The primary purpose of this project is to develop an interactive Tableau-based healthcare analytics platform that transforms patient data into meaningful visual insights for heart disease analysis. The project aims to simplify the exploration of healthcare information by presenting demographic, lifestyle, and clinical indicators through dashboards, KPI cards, interactive filters, and data storytelling.

The solution enables users to identify heart disease trends, compare patient groups, and understand relationships between risk factors such as age, BMI, smoking, diabetes, stroke history, and physical activity. By integrating Tableau with a Flask web application, the project provides an accessible platform for viewing analytical dashboards and stories through a web interface.

This project demonstrates the effective use of business intelligence and data visualization techniques in healthcare, supporting data-driven decision-making, promoting preventive healthcare awareness, and improving the interpretation of complex medical datasets.

2. IDEATION PHASE

The ideation phase focused on understanding the healthcare problem, identifying stakeholder needs, and exploring possible analytical solutions before the development of the Tableau dashboard. Various healthcare perspectives, patient requirements, and visualization approaches were considered to ensure the final solution effectively supports heart disease analysis and data-driven decision-making.

2.1 Problem Statement

Heart disease continues to be one of the leading causes of mortality worldwide, with its prevalence increasing due to factors such as unhealthy lifestyles, obesity, smoking, diabetes, and physical inactivity. Although healthcare organizations collect large volumes of patient information, extracting meaningful insights from this data remains challenging because raw datasets are difficult to interpret without effective visualization.

Traditional data analysis methods often require significant time and technical expertise, making it difficult for healthcare professionals and researchers to quickly identify disease patterns, high-risk patient groups, and relationships between demographic, lifestyle, and clinical factors. As a result, opportunities for early awareness, preventive healthcare, and informed decision-making may be limited.

This project addresses these challenges by developing an interactive **Heart Disease Analysis Dashboard** using Tableau. The solution transforms raw healthcare data into meaningful visual insights through interactive dashboards, KPI cards, filters, calculated fields, and Tableau Story. By integrating the analytical dashboards

into a Flask-based web application, the project provides an accessible platform that enables users to explore healthcare trends, identify risk factors, and support data-driven healthcare analysis.

Figure 2.1: Problem Statement Canvas

Ideation Phase Define the Problem Statements

Date	5 July 2026
Team ID	
Project Name	Heart Disease Analysis with Tableau
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - who are they?	Describe the customer and their attributes here
I'm trying to	List their outcome or "job" the care about - what are they trying to achieve?	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way - what bothers them most?	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists - what needs to be solved?	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view - how does it impact them emotionally?	Describe the emotions the result from experiencing the problems or barriers

Reference: <https://miro.com/templates/customer-problem-statement/>

Example:

I am a traveler	I'm trying to book flights on my phone	but it takes a long time	because this website is not mobile and doesn't have a mobile version	which makes me feel Frustrated
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Customer Problem Statement Template

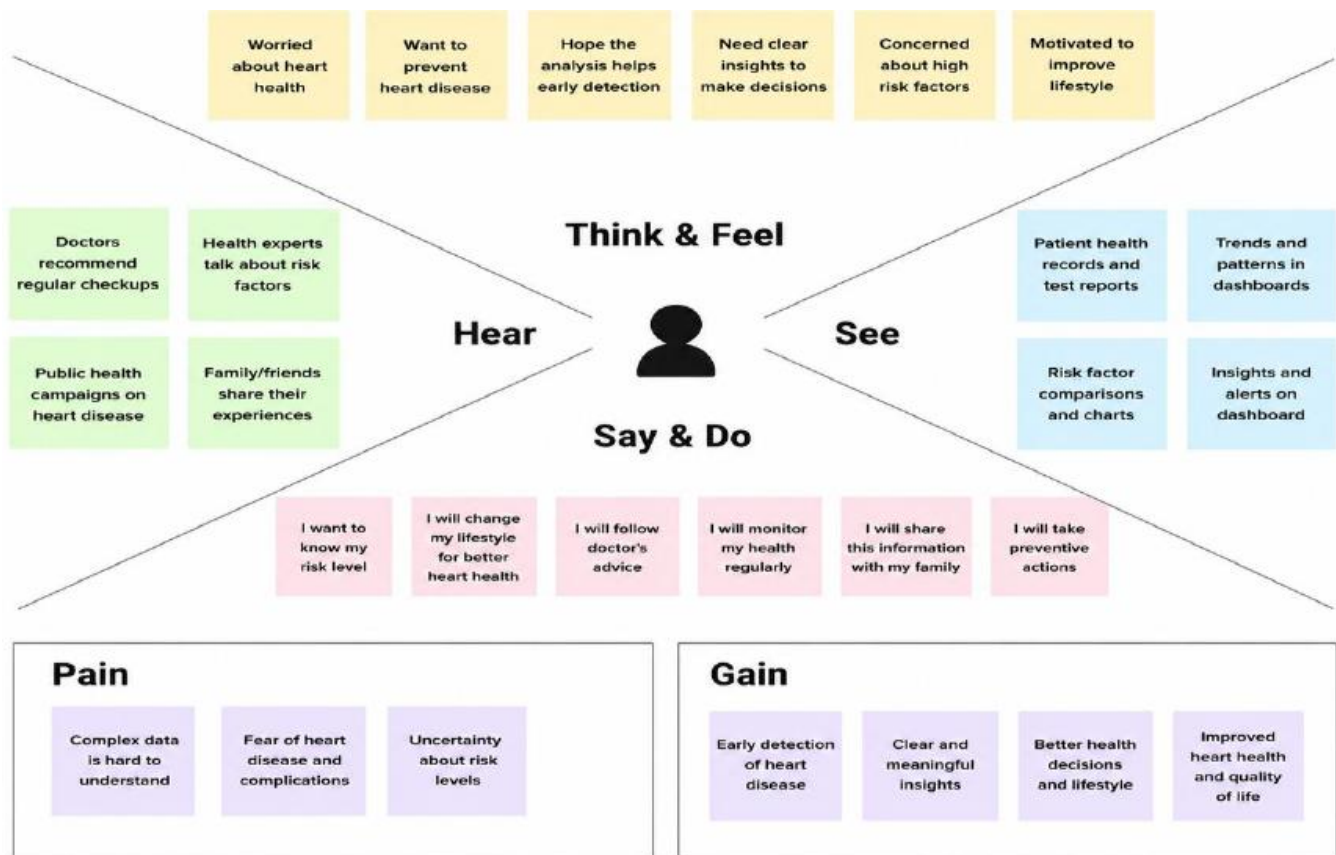
I am	I'm trying to	But	Because	Which makes me feel
I am Cardiologist Health Officer Working Professional	I'm trying to Identify Risk Factors Analyze Patient Data Monitor Health Trends	But Large Healthcare Data Limited Visualizations Hidden Patterns	Because Multiple Risk Factors Traditional Reporting Lack of Insights	Which makes me feel Concerned Challenged Uncertain

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Cardiologist	Identify high-risk patients	Complex health data	Scattered patient records	Concerned about diagnosis
PS-2	Health Officer	Analyze disease trends	Limited visual insights	Limited visual insights	Challenged in planning

2.2 EMPATHY MAP

The Empathy Map Canvas was developed to understand the needs, expectations, challenges, and behaviors of the key stakeholders involved in heart disease analysis. It provides valuable insights into how different users interact with healthcare data and highlights the importance of presenting complex medical information through simple and interactive visualizations. Understanding these perspectives helped in designing a user-centric dashboard that supports efficient analysis and informed decision-making.

Figure 2.2: Empathy Map Canvas



The Empathy Map illustrates the perspectives of the primary stakeholders, including the cardiologist, healthcare administrator, and patient. It captures what they think, observe, say, and do while interacting with healthcare data, along with their challenges and expectations. These insights guided the development of interactive dashboards, KPI cards, filters, and Tableau Story to ensure the solution effectively supports healthcare analysis, risk identification, and data-driven decision-making.

2.2 EMPATHY MAP CANVAS

1. Dr. Sharma – Cardiologist

- **Says:** "Which patient groups are at the highest risk of heart disease? How can early intervention improve patient outcomes?"
- **Thinks:** Needs reliable insights to identify risk factors and support preventive treatment strategies.
- **Does:** Analyzes patient data using Tableau dashboards, comparing age, BMI, diabetes, smoking habits, and cholesterol-related indicators.
- **Feels:** Responsible for improving patient health through timely diagnosis and evidence-based clinical decisions.

Ideation Phase Empathize & Discover

Date	5 July 2026
Team ID	
Project Name	Heart Disease Analysis with Tableau
Maximum Marks	4 Marks

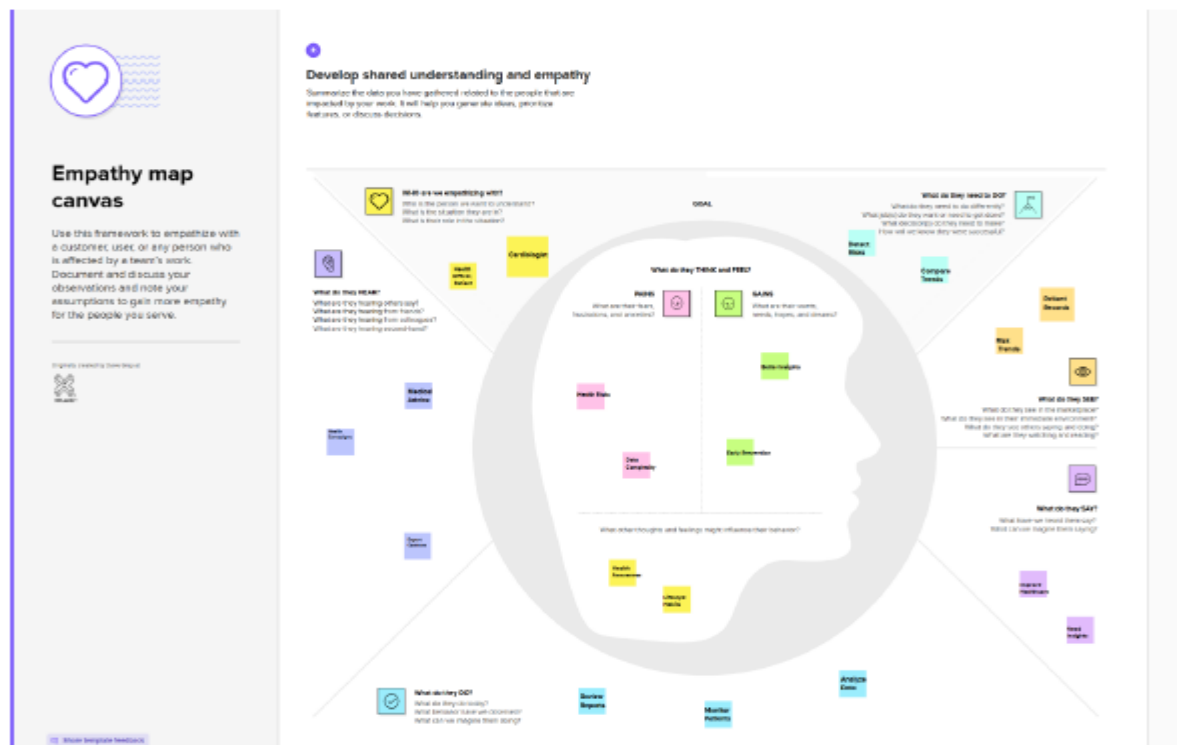
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Reference: <https://www.mural.co/templates/empathy-map-canvas>



2.3 Brainstorming

The brainstorming phase focused on generating innovative ideas and identifying effective approaches for analyzing heart disease data using interactive visualizations. Various healthcare challenges, user requirements, and analytical perspectives were explored to determine the most suitable solution. Techniques such as **How Might We (HMW)**, **Brainstorming Ideas**, **Affinity Mapping**, and **Solution Prioritization** were used to organize ideas and select the final project approach.

Figure 2.3: How Might We (HMW) Statements

Ideation Phase Brainstorm & Idea Prioritization Template

Date	5 July 2026
Team ID	
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Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Template



Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

10 minutes to prepare
1 hour to collaborate
3-6 people recommended

Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

10 minutes

Team gathering

Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

Set the goal

Think about the problem you'll be focusing on solving in the brainstorming session.

Learn how to use the facilitation cards

Use the Facilitation Paper prompts to run a happy and productive session.

[Open article](#)

1

Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

5 minutes

How might we...

How might we use interactive data visualization to identify heart disease risk factors and suggest better healthcare decisions?

2

Key rules of brainstorming

To run an smooth and productive session

- Stay in topic
- Encourage wild ideas
- Defer judgment
- Listen to others
- Go for volume
- If possible, be visual

Step-2: Brainstorm, Idea Listing and Grouping

2 Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

Tip: Use sticky notes, color, and arrows to group ideas.

Person 1

How might we organize healthcare data for accurate analysis?

How might we analyze patient demographics to understand disease patterns?

How might we identify the major risk factors associated with heart disease?

Person 2

How might we analyze lifestyle factors to identify high-risk individuals?

How might we evaluate the relationship between physical activity and heart health?

How might we measure the impact of smoking on heart disease?

Person 3

How might we use data to explore disease patterns?

How might we simplify heart disease analysis using effective visualizations?

How might we present complex healthcare data through interactive dashboards?

Person 4

How might we compare different patient groups to reveal hidden trends?

How might we highlight meaningful insights from heart disease data?

How might we combine multiple analysis findings to discover anything?

Person 5

How might we use data to improve healthcare risk through interactive analysis?

How might we increase awareness of heart disease risk through interactive analysis?

How might we support healthcare professionals in making data-driven decisions?

Group Ideas

Take turns sharing your ideas while clustering similar or related notes to your group. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

10 minutes

Tip: Give constructive feedback to others. Focus on the ideas, not the person.

Group 1

- Analyze demographic factors (Age, Gender, Race)
- Study lifestyle risk factors (Smoking, Alcohol, Physical Activity)
- Identify high-risk patient groups

Group 2

- Develop interactive Tableau dashboards and stories
- Generate healthcare insights for better decision-making

Step-3: Idea Prioritization

3 Prioritize

You're likely to have a lot of ideas. Prioritize them by importance and feasibility. Place your ideas on the grid to determine which ideas are important and which are feasible.

20 minutes

Tip: The ideas you place in the top-left quadrant are the most important and feasible. The ideas you place in the bottom-right quadrant are the least important and least feasible.

After you collaborate

You can export the results as an image or PDF for a future meeting with stakeholders or for a future meeting with stakeholders.

Quick actions

- Share the board: Share a link to the board with stakeholders to see their thoughts on the ideas.
- Export Results: Export a copy of the board as a PDF or CSV to share with stakeholders.

Keep moving forward

- Strategic roadmap: Define the roadmap of the project or initiative.
- Customer experience roadmap: Understand customer needs, behaviors, and expectations for an experience.
- Strength, weaknesses, opportunities & threats: Identify strengths, weaknesses, opportunities, and threats (SWOT) to develop a plan.

Conclusion

The brainstorming process helped identify the most practical and impactful solution for heart disease analysis. After evaluating multiple ideas, the team selected an interactive Tableau-based dashboard integrated with a Flask web application. This approach enables users to analyze demographic, lifestyle, and clinical factors through intuitive visualizations, supporting better understanding of heart disease patterns and informed healthcare decision-making.

3. REQUIREMENT ANALYSIS

The Requirement Analysis phase focused on identifying stakeholder needs, defining the functional and non-functional requirements of the system, and designing the overall workflow for the Heart Disease Analysis project. This phase ensured that the proposed solution effectively addresses the problem statement while providing an intuitive, interactive, and user-friendly analytical platform for healthcare data visualization.

3.1 Customer Journey Map

The Customer Journey Map illustrates the step-by-step interaction of a user while exploring the Heart Disease Analysis platform. It highlights the user's objectives, touchpoints, challenges, and opportunities at each stage of the journey. This analysis helped in designing an intuitive dashboard and web application that simplifies healthcare data exploration and improves the overall user experience.

Figure 3.1: Customer Journey Map



The customer journey demonstrates how users progress from understanding the purpose of the application to exploring dashboards, analyzing healthcare insights, interacting with filters, and interpreting Tableau Story findings. Identifying user pain points during this process enabled the development of a solution that provides

clear visualizations, interactive analytics, and easy navigation for healthcare professionals, researchers, and patients.

3.2 SOLUTION REQUIREMENT

The Heart Disease Analysis project was designed to transform raw healthcare data into meaningful visual insights through interactive dashboards and a web-based interface. The solution requirements were identified to ensure that the application provides effective data visualization, user-friendly interaction, and reliable analytical capabilities for healthcare professionals, researchers, and patients.

Project Design Phase-II Solution Requirements (Functional & Non-functional)

Date	5 July 2026
Team ID	
Project Name	Heart Disease Analysis with Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Home Page	Display project overview and introduction. Provide navigation to all project sections. Responsive homepage layout.
FR-2	Dashboard Analysis	Display interactive Tableau Dashboard. Visualize KPIs, charts, and graphs. Support interactive filtering and exploration.
FR-3	Data Story	Display Tableau Story. Present analytical insights sequentially. Enable users to navigate story points.
FR-4	Project Information	Display About, Objectives, Technologies, Developer, and Contact sections. Provide project documentation and insights. Support smooth navigation between sections.

Non-Functional Requirements

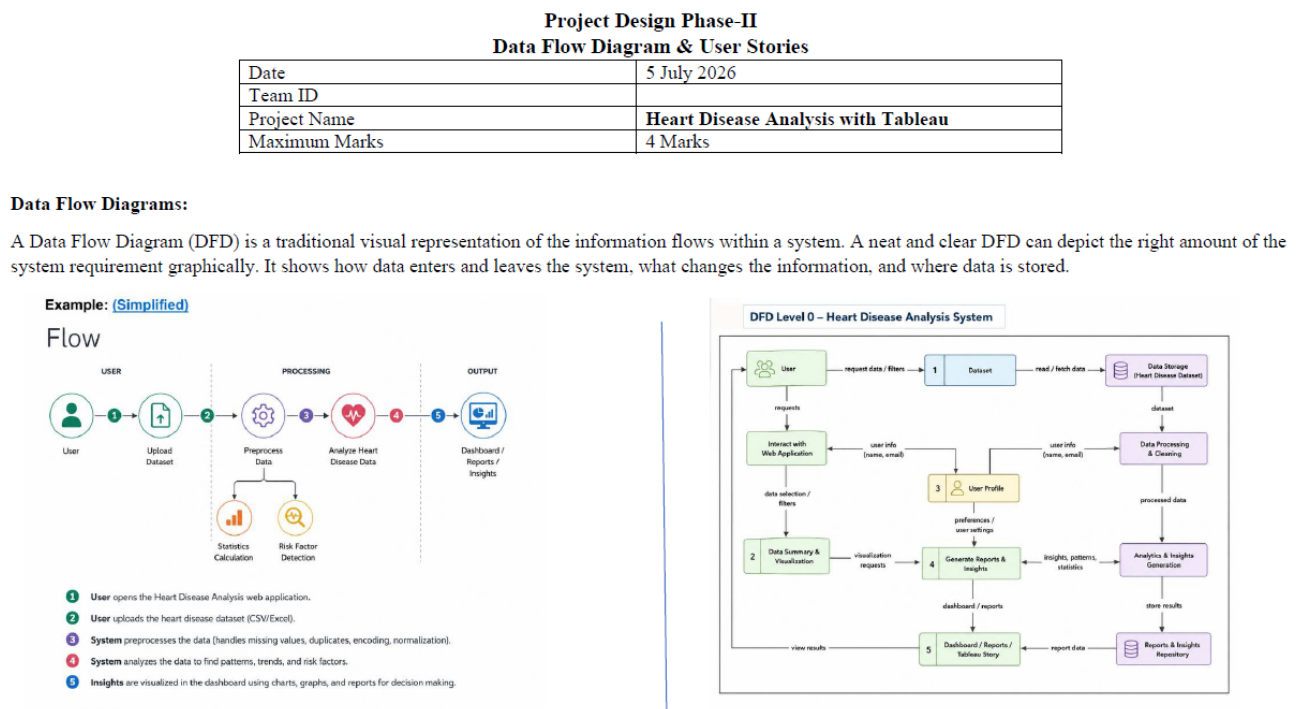
FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application shall provide an intuitive, responsive, and user-friendly interface for easy navigation.
NFR-2	Security	The application shall securely embed Tableau visualizations and validate user interactions without exposing sensitive information.
NFR-3	Reliability	The application shall consistently display dashboards, stories, and project information without failures.
NFR-4	Performance	Pages, dashboards, and visualizations shall load efficiently with minimal response time.
NFR-5	Availability	The web application and published Tableau dashboards shall be accessible whenever the hosting services are available.

NFR-6	Scalability	The application shall support future enhancements such as additional dashboards, datasets, and analytical modules without major architectural changes.
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3.3 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) illustrates the movement of data throughout the Heart Disease Analysis system. It represents how healthcare data is collected, processed, visualized, and presented to users through interactive dashboards and a web-based interface. The DFD provides a clear understanding of the relationship between the dataset, Tableau analytics, Flask application, and end users.

Figure 3.2: Data Flow Diagram



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Visitor (Web User)	Home Page	USN-1	As a visitor, I want to access the home page to understand the project overview.	Home page loads successfully with project introduction.	High	Sprint-1
Visitor (Web User)	Project Information	USN-2	As a visitor, I want to view the project objectives and workflow.	Objectives and workflow are displayed correctly.	High	Sprint-1
Visitor (Web User)	Dashboard	USN-3	As a visitor, I want to view the interactive Tableau dashboard.	Dashboard loads and supports interaction.	High	Sprint-1
Visitor (Web User)	Tableau Story	USN-4	As a visitor, I want to explore the Tableau Story for detailed analysis.	Story loads with all story points accessible.	High	Sprint-1
Visitor (Web User)	Insights	USN-5	As a visitor, I want to view the key findings from the analysis.	Insights section displays correctly.	Medium	Sprint-1
Visitor (Web User)	Technology Stack	USN-6	As a visitor, I want to know the technologies used in the project.	Technologies section is displayed.	Low	Sprint-1
Healthcare Professional	Risk Analysis	USN-7	As a healthcare professional, I want to analyze heart disease trends for better clinical decisions.	Dashboard provides meaningful visual insights.	High	Sprint-2
Healthcare Professional	Data Filtering	USN-8	As a healthcare professional, I want to filter data by demographic and lifestyle factors.	Tableau filters work correctly.	High	Sprint-2
Researcher	Trend Analysis	USN-9	As a researcher, I want to compare demographic and health patterns.	Charts update based on selected filters.	Medium	Sprint-2
Public Health Officer	Decision Support	USN-10	As a public health officer, I want to identify high-risk groups for planning awareness programs.	Dashboard highlights risk patterns clearly.	High	Sprint-2

Administrator	Website Management	USN-11	As an administrator, I want the application pages to load without errors.	All pages function correctly.	Medium	Sprint-3
Administrator	Dashboard Maintenance	USN-12	As an administrator, I want to update the embedded Tableau dashboard when new versions are published.	Updated Tableau content is displayed successfully.	Medium	Sprint-3

3.4 TECHNOLOGY STACK

The Heart Disease Analysis project utilizes a combination of business intelligence, web development, and visualization technologies to transform raw healthcare data into interactive analytical dashboards. Tableau is used for data visualization and storytelling, while Flask provides a lightweight web framework to integrate the dashboards into a user-friendly application. The selected technologies ensure efficient data analysis, accessibility, and an enhanced user experience.

Table 3.1: Technology Stack

**Project Design Phase-II
Technology Stack (Architecture & Stack)**

Date	5 July 2026
Team ID	
Project Name	Heart Disease Analysis with Tableau
Maximum Marks	4 Marks

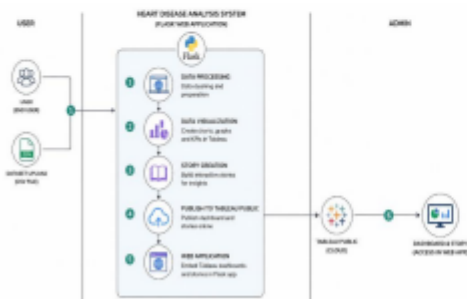
Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Heart Disease Analysis with Tableau:



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	Data Source	Heart disease dataset containing patient demographics, lifestyle, and clinical information.	CSV Dataset
2	Data Preparation	Cleaning, formatting, and preprocessing of the dataset before visualization.	Tableau Desktop
3	Calculated Fields	Creation of custom measures and KPIs for enhanced analysis.	Tableau Calculated Fields
4	Data Visualization	Development of charts, graphs, and visual analytics.	Tableau Desktop
5	Dashboard Development	Creation of an interactive dashboard with filters and KPI cards.	Tableau Dashboard
6	Story Development	Presentation of analytical insights using sequential story points.	Tableau Story
7	Dashboard Publishing	Publishing dashboards and stories for online accessibility.	Tableau Public
8	User Interface	Responsive web interface to access dashboards, stories, project information, and developer details.	HTML5, CSS3, Bootstrap, JavaScript
9	Application Logic	Handles webpage routing and embeds Tableau Dashboard and Story within the application.	Python, Flask
10	File Storage	Stores datasets, images, and project assets locally.	Local File System
11	Infrastructure (Server)	Local deployment and execution of the web application.	Flask Development Server

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source frameworks and tools used to develop the application.	Flask, Bootstrap, Tableau Public

2	Security Implementations	Client-side validation, secure routing, and safe embedding of Tableau visualizations.	HTML5 Validation, Flask Routing
3	Scalable Architecture	Modular architecture separating data, visualization, and presentation layers.	Three-Tier Architecture (Data Layer, Visualization Layer, Presentation Layer)
4	Availability	Dashboard and Story published online and accessible through the web application.	Tableau Public, Flask
5	Performance	Optimized dashboards, responsive frontend, and lightweight web application for faster loading.	Tableau Public, Bootstrap, JavaScript

Technology Description

The project combines Tableau Desktop for interactive dashboard development and Tableau Public for publishing dashboards and stories. Flask serves as the backend framework, enabling seamless integration of Tableau visualizations into a web application. HTML, CSS, Bootstrap, and JavaScript are used to build a responsive and user-friendly interface, while Python handles backend functionality. GitHub is utilized for version control and project management, and Visual Studio Code serves as the primary development environment.

4. PROJECT DESIGN

The Project Design phase focuses on transforming the identified requirements into a practical and effective solution. Based on the problem statement and stakeholder needs, an interactive Tableau-based analytics platform was designed to simplify healthcare data analysis through dashboards, visualizations, KPI cards, and web integration. This phase defines how the proposed solution addresses the challenges identified during the ideation and requirement analysis stages.

4.1 PROBLEM SOLUTION FIT

The Heart Disease Analysis project addresses the challenge of interpreting complex healthcare datasets by providing an interactive business intelligence solution using Tableau. Traditional methods of analyzing patient records often involve manual interpretation of large datasets, making it difficult to identify meaningful trends and relationships. The proposed solution transforms raw healthcare data into interactive dashboards that enable users to analyze demographic, lifestyle, and clinical factors efficiently.

The solution incorporates **three dashboards, ten interactive visualizations, KPI cards, calculated fields, interactive filters, and Tableau Story** to present healthcare information in a structured and user-friendly manner. By integrating these visualizations into a Flask-based web application, the project improves accessibility and enables healthcare professionals, researchers, and patients to explore analytical insights through an intuitive interface.

Figure 4.1: Problem Solution Fit

Project Design Phase
Problem – Solution Fit Template

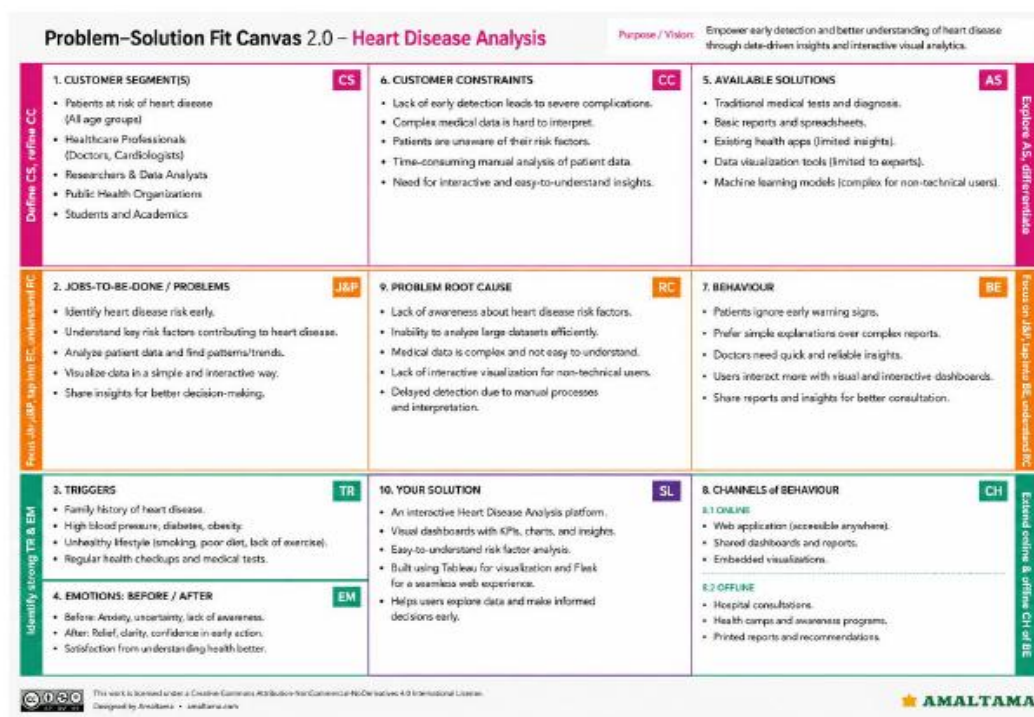
Date	6 July 2026
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Project Name	Heart Disease Analysis with Tableau
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ Understand the existing situation in order to improve it for your target group.



The proposed solution successfully aligns with the identified problem by converting healthcare data into meaningful visual insights that support trend analysis, risk identification, and informed decision-making. The combination of interactive dashboards and web integration ensures that users can easily explore healthcare information without requiring advanced technical knowledge.

4.2 PROPOSED SOLUTION

The proposed solution is an interactive **Heart Disease Analysis platform** developed using Tableau and integrated into a Flask web application. The solution consolidates patient demographic information, lifestyle factors, and medical conditions into a centralized analytical environment where users can explore healthcare trends through interactive dashboards and visualizations.

The platform consists of **three analytical dashboards, ten visualizations, KPI cards, interactive filters,** and a **Tableau Story** that collectively provide a comprehensive understanding of heart disease patterns. Users can compare patient groups, analyze relationships between health indicators, and explore risk factors such as age, BMI, smoking habits, diabetes, stroke history, and physical activity.

Project Design Phase Proposed Solution Template

Date	6 July 2026
Team ID	
Project Name	Heart Disease Analysis with Tableau
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Heart disease is one of the leading causes of mortality worldwide. Analyzing large healthcare datasets to identify risk factors and disease patterns is complex. There is a need for an interactive platform that transforms raw health data into meaningful visual insights to support early detection, preventive care, and informed healthcare decisions.
2	Idea / Solution Description	Develop a Heart Disease Analysis web application using Tableau and Flask that visualizes healthcare data through interactive dashboards, KPI cards, charts, and Tableau Stories. The application enables users to explore disease trends, demographic patterns, lifestyle risk factors, and analytical insights through an intuitive web interface.
3	Novelty / Uniqueness	The solution combines Tableau's interactive data visualization and storytelling capabilities with a Flask-based web application, providing a centralized platform for exploring heart disease data through dashboards, stories, and insights in an easy-to-understand format.
4	Social Impact / Customer Satisfaction	The application increases awareness of heart disease risk factors, supports healthcare professionals in data-driven decision-making, and helps researchers, students, and the general public understand disease trends through interactive visual analytics, contributing to preventive healthcare and better public awareness.
5	Business Model (Revenue Model)	The solution can be offered as a subscription-based healthcare analytics platform for hospitals, clinics, research organizations, and educational institutions. Additional revenue can be generated through premium analytical dashboards, customized reporting services, institutional licensing, and healthcare consultancy solutions.
6	Scalability of the Solution	The application is designed with a modular architecture that supports the integration of additional healthcare datasets, advanced dashboards, predictive analytics modules, cloud deployment, authentication, role-based access, and real-time healthcare data sources, making it suitable for future expansion.

To improve accessibility, the Tableau dashboards and story are embedded within a Flask-based web application that provides dedicated pages for project information, dashboards, insights, technologies, developer details, and contact information. This integration enables users to access healthcare analytics through a centralized and user-friendly platform.

4.3 SOLUTION ARCHITECTURE

The Solution Architecture illustrates the overall structure of the Heart Disease Analysis system and the interaction between its major components. The architecture begins with the Heart Disease dataset, which serves as the primary data source. The dataset is processed in Tableau Desktop, where interactive dashboards, KPI cards, calculated fields, and Tableau Story are developed.

The published dashboards are hosted on Tableau Public and integrated into the Flask web application. Users access the application through a web browser, where they can interact with dashboards, apply filters, explore stories, and analyze healthcare insights through an intuitive interface. This architecture ensures efficient data visualization, seamless integration, and improved accessibility while maintaining a simple and scalable system design.

Figure 4.2: Solution Architecture

Project Design Phase Solution Architecture	
Date	6 Jul 2026
Team ID	
Project Name	Heart Disease Analysis with Tableau
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

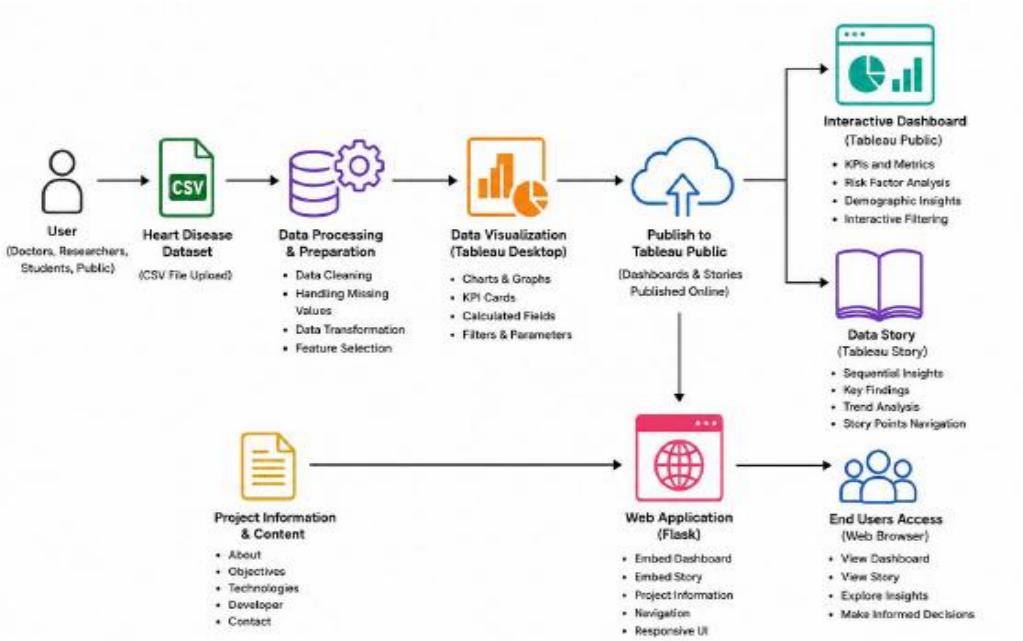


Figure 1: Architecture and data flow of the Heart Disease Analysis application

The architecture demonstrates a clear flow of information from the healthcare dataset to the end user. By combining Tableau for visualization and Flask for web integration, the solution provides a lightweight yet effective platform for healthcare analytics and interactive decision support.

5. PROJECT PLANNING & SCHEDULING

The Project Planning and Scheduling phase outlines the systematic execution of the Heart Disease Analysis project. The development process was divided into multiple sprints, allowing the team to complete data

preparation, dashboard development, web integration, testing, and documentation in an organized manner. Sprint-based planning ensured effective task management, timely completion, and continuous progress throughout the project lifecycle.

5.1 Project Planning

The project was planned using an iterative sprint-based approach to ensure the systematic completion of all development activities. Each sprint focused on a specific set of objectives, including data preprocessing, dashboard development, KPI creation, Tableau Story design, Flask web integration, testing, and documentation. This structured planning approach improved project organization, enabled continuous monitoring of progress, and ensured timely completion of all deliverables.

Table 5.1: Product Backlog & Sprint Planning

Project Planning Phase	
Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)	
Date	6 July 2026
Team ID	
Project Name	Heart Disease Analysis with Tableau
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule & Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Preparation	USN-1	Collect and prepare the heart disease dataset for analysis.	3	High	Team
Sprint-1	Data Cleaning	USN-2	Clean missing values and prepare data for visualization.	2	High	Team
Sprint-1	Data Visualization	USN-3	Create charts and graphs using Tableau.	5	High	Team
Sprint-2	Dashboard Development	USN-4	Develop an interactive Tableau Dashboard with KPI cards and filters.	8	High	Team
Sprint-2	Story Development	USN-5	Create Tableau Story with sequential analytical insights.	5	High	Team
Sprint-3	Web Application	USN-6	Develop Flask web application with responsive pages.	8	High	Team
Sprint-3	Dashboard Integration	USN-7	Embed Tableau Dashboard into the Flask application.	5	High	Team
Sprint-3	Story Integration	USN-8	Embed Tableau Story into the Flask application.	3	Medium	Team
Sprint-4	Project Documentation	USN-9	Prepare project documentation, reports, and diagrams.	5	High	Team
Sprint-4	Testing & Deployment	USN-10	Test the application and perform final deployment.	5	High	Team

Table 5.3: Velocity & Burndown

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	20	2 Days	01 Jul 2026	02 Jul 2026	20	02 Jul 2026
Sprint-2	20	2 Days	03 Jul 2026	04 Jul 2026	20	04 Jul 2026
Sprint-3	20	1 Day	05 Jul 2026	05 Jul 2026	20	05 Jul 2026
Sprint-4	20	2 Days	06 Jul 2026	07 Jul 2026	20	07 Jul 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

Day	Remaining Story Points
Day 1	20
Day 2	18
Day 3	15
Day 4	12
Day 5	9
Day 6	6
Day 7	4
Day 8	2
Day 9	1
Day 10	0

Project Planning Summary

The project was executed through multiple development sprints, beginning with dataset preparation and requirement analysis, followed by dashboard design, visualization development, calculated field implementation, web integration, testing, and final documentation. The sprint-based methodology enabled efficient task allocation, continuous progress tracking, and successful completion of the project within the planned schedule.

6. FUNCTIONAL AND PERFORMANCE TESTING

The Functional and Performance Testing phase was conducted to verify that the Heart Disease Analysis application performs as expected under normal operating conditions. The testing process evaluated the functionality of the Tableau dashboards, web application, interactive filters, KPI cards, and navigation features to ensure accuracy, usability, and reliability. The results confirmed that the application successfully met the intended project requirements.

6.1 Functional Testing

Functional testing was performed to validate the core features of the Heart Disease Analysis application. Each module was tested to ensure that it functioned correctly and produced the expected results. The testing included dashboard loading, navigation, KPI display, filter interaction, Tableau Story accessibility, and web page responsiveness.

Table 6.1: Functional Testing

Test Case	Expected Result	Actual Result	Status
Home Page Loading	Home page opens successfully	Loaded Successfully	Pass
Dashboard Loading	Tableau dashboard loads correctly	Loaded Successfully	Pass
Story Loading	Tableau Story opens successfully	Loaded Successfully	Pass
KPI Cards Display	KPI values displayed correctly	Displayed Successfully	Pass
Dashboard Filters	Filters update visualizations correctly	Working Successfully	Pass
Navigation Menu	All pages navigate correctly	Working Successfully	Pass
About & Developer Page	Information displayed correctly	Displayed Successfully	Pass
Contact Page	Contact details accessible	Displayed Successfully	Pass

Functional Testing Summary

The functional testing results indicate that all major features of the application operate as expected. The dashboards, stories, filters, KPI cards, navigation components, and web pages function correctly without any observed issues.

6.2 Performance Testing

Performance testing was conducted to evaluate the responsiveness and usability of the application during normal usage. The assessment focused on dashboard loading, filter responsiveness, visualization rendering, and overall application performance within the Flask web environment.

Table 6.2: Performance Testing

Project Performance Test

Date	7 July 2026
Team ID	
Project Name	Heart Disease Analysis with Tableau
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1	Data Rendered	4,500 Records Loaded
2	Data Preprocessing	Data Cleaning & Formatting Completed
3	Utilization of Filters	7 Interactive Filters Applied
4	Calculation Fields Used	3 Calculated Fields Created
5	Dashboard Design	10 Visualizations, 4 KPI Cards
6	Story Design	6 Story Points, 10 Visualizations

Performance Testing Summary

The application demonstrated stable performance during testing. Dashboards and stories loaded successfully, filters responded accurately, and all visual components were rendered correctly. The Flask web application provided a smooth and responsive user experience throughout the testing process.

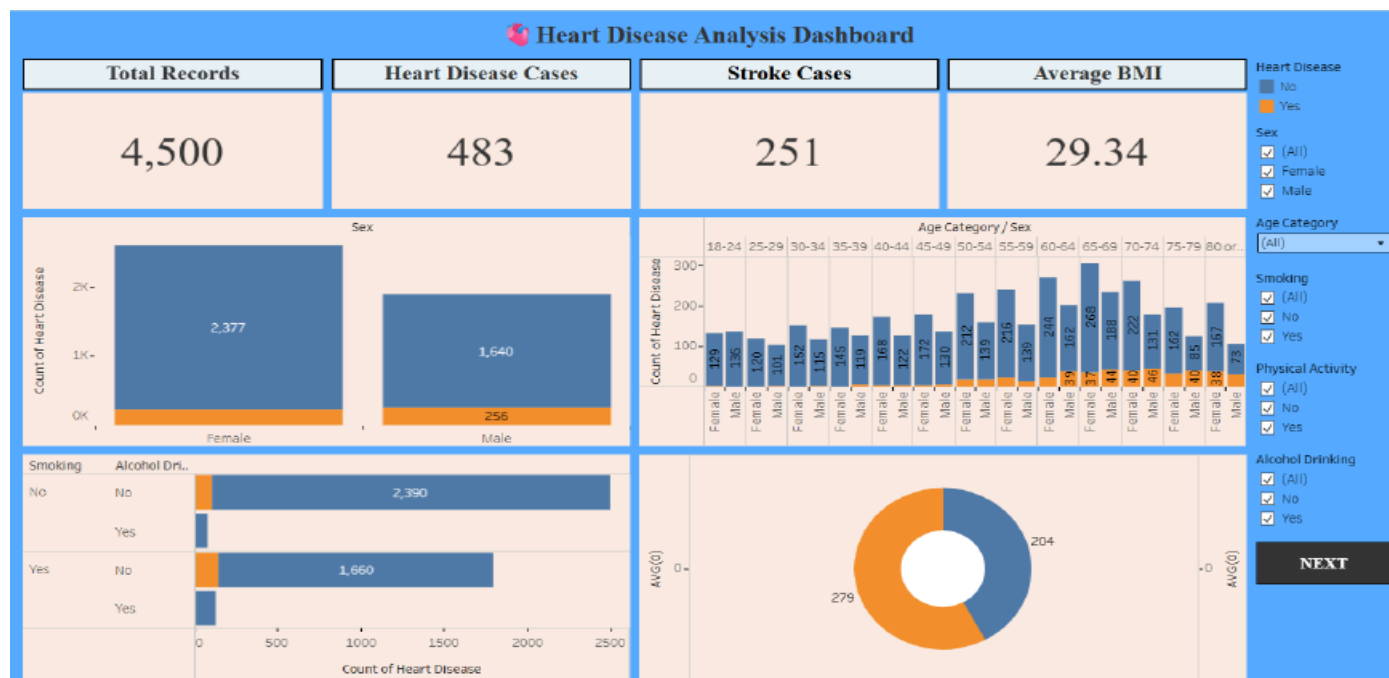
7. DOCUMENTATION & DEMO

7.1 Final Dashboard

The final Tableau dashboard presents a comprehensive analysis of heart disease by combining demographic, lifestyle, and clinical indicators into an interactive visualization platform. The dashboard enables users to explore healthcare trends, compare patient groups, and identify major risk factors through KPI cards,

interactive charts, calculated fields, and dynamic filters. The centralized dashboard provides meaningful insights that support healthcare awareness and informed decision-making.

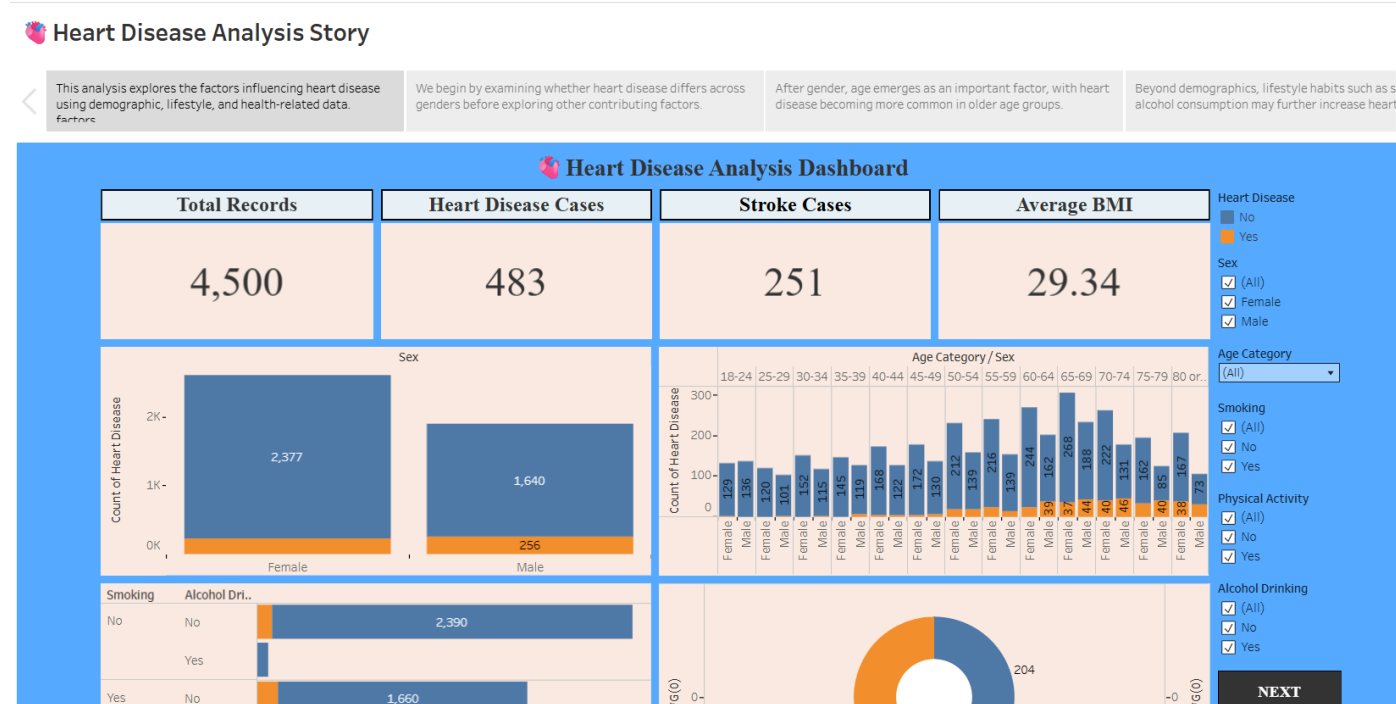
Figure 7.1: Final Tableau Dashboard



7.2 Tableau Story

The Tableau Story presents the analytical findings in a structured sequence, guiding users through important healthcare insights derived from the dataset. Each story point focuses on a different aspect of heart disease analysis, enabling users to understand demographic patterns, lifestyle influences, medical conditions, and overall healthcare trends through an intuitive storytelling approach.

Figure 7.2: Tableau Story



7.3 GitHub Repository

The complete project source code, website resources, and supporting files are maintained using GitHub for version control and collaborative development. The repository ensures organized project management, simplified maintenance, and easy access to the implementation files.

GITHUB REPOSITORY

Web Application Repository URL:

<https://github.com/thiragatisuryasri-analyst/Heart-Disease-Analysis-with-Tableau>

7.5 Tableau Public Links

Dashboard:

https://public.tableau.com/views/HeartDiseaseAnalysis-1_17830862018790/Home?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Tableau Story:

https://public.tableau.com/views/HeartDiseaseAnalysis-2_17832385228620/Story?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7.6 Project Summary

The Heart Disease Analysis project successfully transformed raw healthcare data into an interactive analytical platform using Tableau and Flask. Through three dashboards, ten visualizations, KPI cards, calculated fields, interactive filters, and Tableau Story, the project enables users to analyze heart disease trends, identify risk factors, and explore healthcare insights efficiently. The integration of Tableau with a Flask web application enhances accessibility and provides a centralized platform for healthcare analytics, supporting data-driven awareness and decision-making.

8. ADVANTAGES

Advantages:

- Interactive dashboards simplify the analysis of complex healthcare data.
- KPI cards provide a quick overview of important healthcare indicators.
- Interactive filters enable users to explore specific patient groups dynamically.
- Tableau Story presents analytical findings in a structured and easy-to-understand format.

9. CONCLUSION

The **Heart Disease Analysis using Tableau** project successfully transformed raw healthcare data into an interactive analytical platform that supports meaningful exploration of heart disease trends and associated risk factors. By utilizing Tableau dashboards, KPI cards, calculated fields, interactive filters, and Tableau Story,

the project enables users to analyze demographic, lifestyle, and clinical indicators through intuitive visualizations.

10. FUTURE SCOPE

The current project can be enhanced further by incorporating advanced healthcare analytics and additional interactive features. Future improvements may include:

- Integration of real-time healthcare datasets for continuous analysis.
- Development of predictive models for early heart disease risk assessment.
- Addition of personalized health recommendations based on patient profiles.
- Deployment of the application on cloud platforms for wider accessibility.
- Development of mobile-responsive and cross-platform versions of the application.
- Integration with hospital information systems and electronic health records.
- Expansion of analytical dashboards using larger and more diverse healthcare datasets.